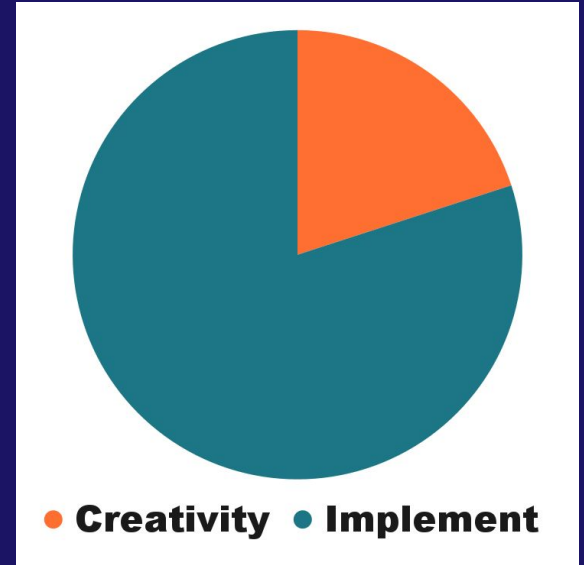




Wuhudsm Contest

By Arpa

A. Explosive String



A. Explosive String

- Replace all '?' with 'L' or replace all '?' with 'R'.

B. Doors



- Greedy
- Std::Set



● **Creativity** ● **Knowledge**
● **Implementation**

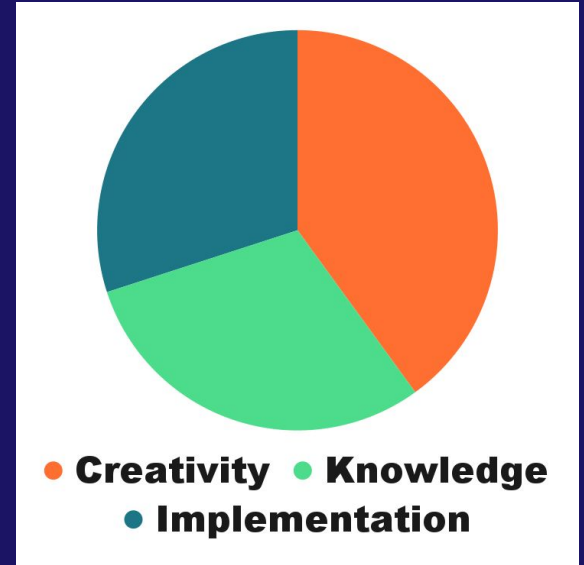
B. Doors

- Greedy.
- Go as far as you can.
- Stop when you can't pass. Then find the maximum c ever seen and change it to zero.
- Maintain c 's in a set or priority queue.

C. Minimum Inversions



- Ordered set



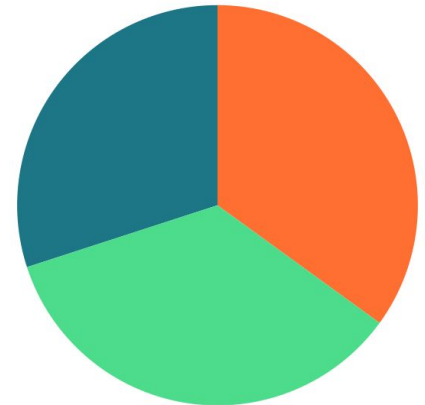
C. Minimum Inversions

- $\text{left}[i]$: number of inversions made by $a[i]$ if we insert it to front.
- $\text{right}[i]$: number of inversions made by $a[i]$ if we insert it to back.
- Calculate $\text{left}[i]$ and $\text{right}[i]$ for each i independently.
- perm : Sort i 's by $\text{left}[i] - \text{right}[i]$
- Answer to j 's query is $\text{sum}(\text{left}[\text{perm}[:j]]) + \text{sum}(\text{right}[\text{perm}[j:]])$

D. Breezy GCD Problem



- Factorization



● **Creativity** ● **Knowledge**
● **Implementation**

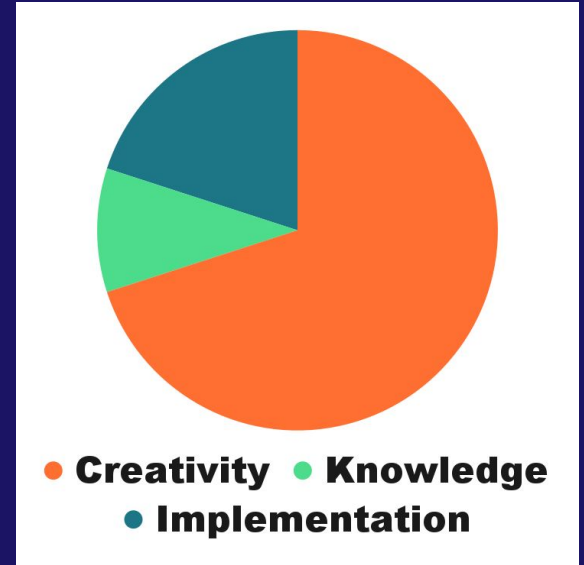
D. Breezy GCD Problem

- If for all i , $l_i < r_i$, output all even numbers.
- Otherwise, there is an $X = l_i = r_i$.
- Factorize X into its prime divisors like p .
- Make sure for every i , there is at least a multiplier of p in $[l_i, r_i]$.

E. Manhattan Tree



- Combinatorics



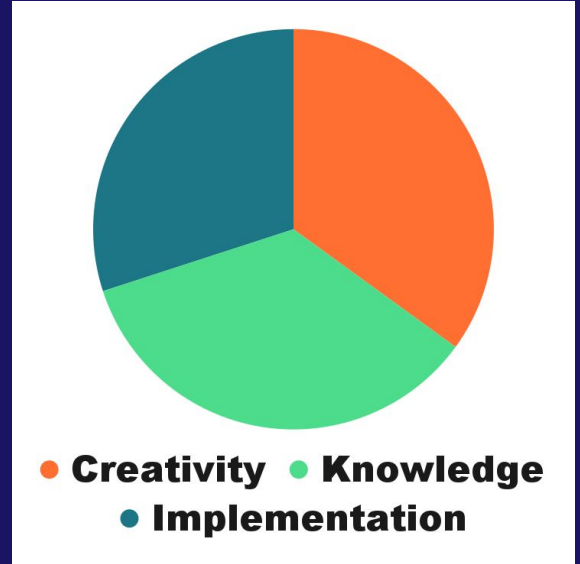
E. Manhattan Tree

- There are only 4 cases:
- One 4-degree vertex and all others are ≤ 2
- Two 3-degree vertices and all others are ≤ 2
- One 3-degree vertices and all others are ≤ 2
- A chain

F. Cool Operations



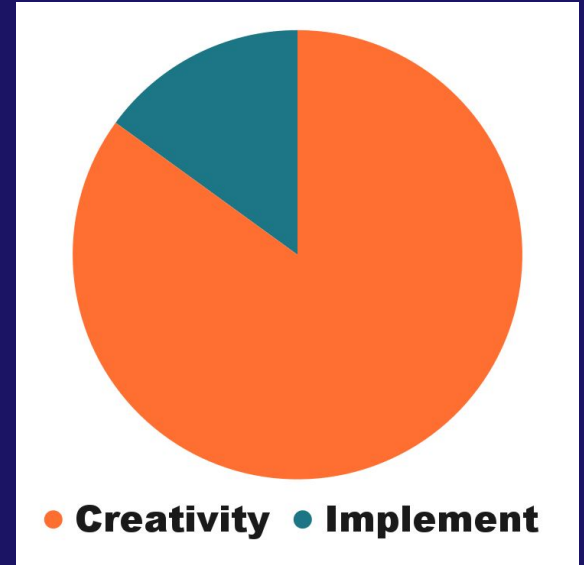
- Sparse Table



F. Cool Operations

- Only the last operation of type 2 matters.
- So the answer is $x + \max(b[j] + a[j+1:r[i]])$.
- There are two other cases: x and $x + \text{sum}(a[l[i]:r[i]])$.
- Let's keep $c[i] = b[i] + a[i+1:]$.
- Build sparse table on c .

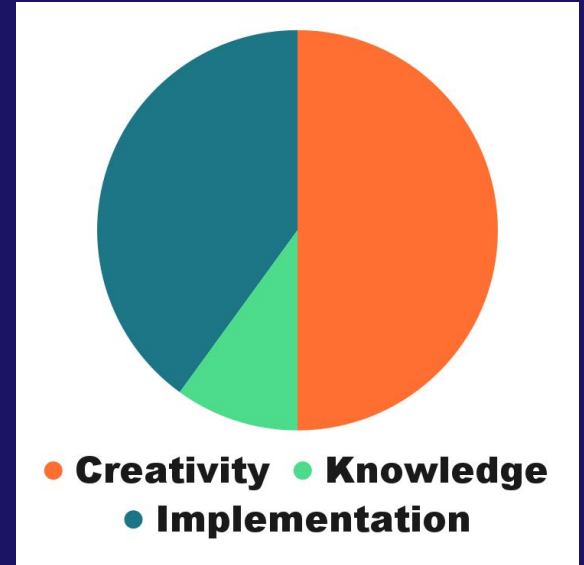
G. Yet Yet Another Binary String Problem



G. Yet Yet Another Binary String Problem

- In these kind of problems you need to take care of 0 block and 1 blocks.
- No block can be destroyed.
- A block can get larger or smaller.
- If the number of blocks and the value of first block (either 0 or 1) matches, a can be converted to b.

H. Mega Polynomial

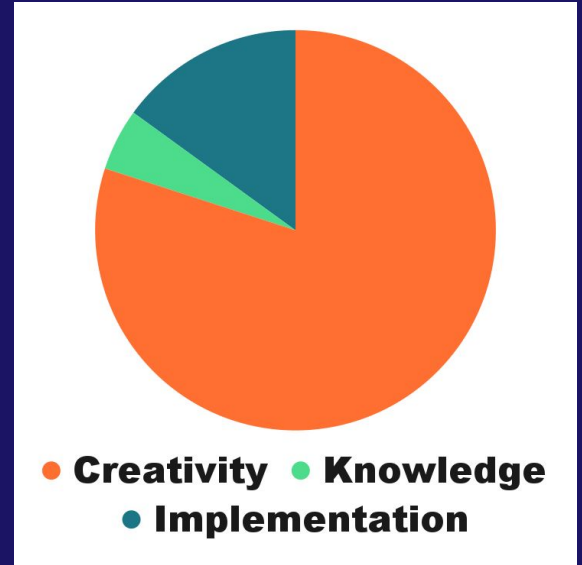


H. Mega Polynomial

- $g^{(k)}(x) = Kx^{n-k-1}(Ax + B) = KAx^{n-k} + KBx^{n-k-1}$.
- $KA = C * n * (n - 1) * \dots * (n - k + 1)$
- $KB = D * (n - 1) * \dots * (n - k)$
- $K = C * n * (n - 1) * \dots * (n - k + 1) / A = D * (n - 1) * \dots * (n - k) / B$
- $BCn = AD * (n - k)$.
- We need to check if $K = C * n * (n - 1) * \dots * (n - k + 1) / A$ is an integer.
- For each prime factor of A, check if its power in numerator is not less than denominator.

I. PermXor

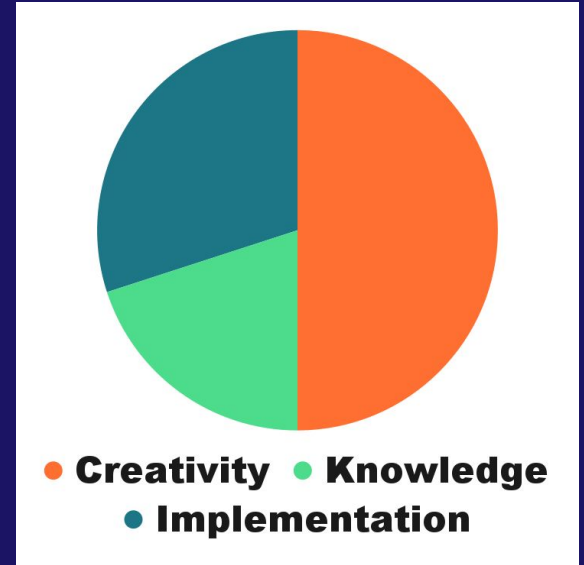
- Constructive



I. PermXor

- Xor of two neighbor is at least 1.
- At least $\text{ceil}(n/2) - 1$ of neighbors has $\text{xor} \geq 2$.
- At least $\text{ceil}(n/2^k) - 1$ of neighbors has $\text{xor} \geq 2^k$.
- This is grey encoding.
- Xor of adjacent things follow this pattern: 1,2,1,4,1,2,1,8,...
- Set last element to $n - 1$ and other elements will be determined.

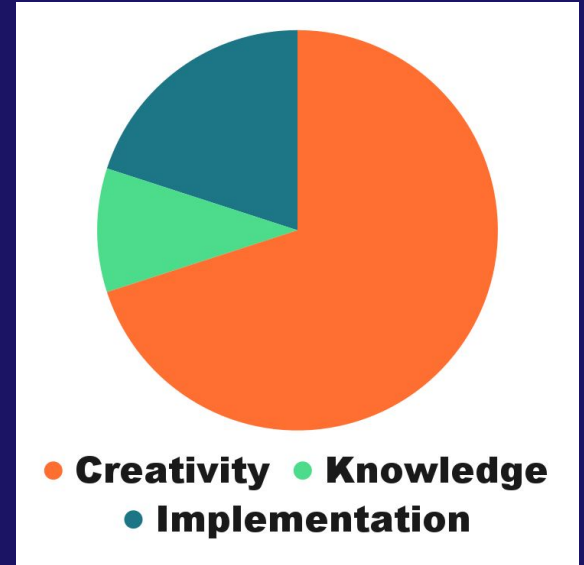
J. Equal Node Sum



J. Equal Node Sum

- (the number of non-leaf nodes) * (min degree of non-leaf nodes) = $O(n)$
- $dp[v][j][k=0/1][m=0/1]$
 - v : the node.
 - j : the sum value of every nodes in subtree v .
 - k : the weight of node v .
 - m : the weight of v 's parent.
- $dp[v][j][k][m] = \sum dp[son[v][x]][j][t[x]][k]$. Here $\sum t[x] = j - k - m$.
- We skip leaf nodes in DP calculation.
- The complexity is $O(n \sqrt{n})$.

K. Mad MAD Sum II



K. Mad MAD Sum II

- MADDES array = $[n, n, \dots, n]$: $n * n * (n - 1) / 2$
- Second MADDES array = $[n, n, \dots, n, 0]$: $n * n * (n - 1) / 2 - n$
- For all m in range $[0, n * n * (n - 1) / 2 - n]$ answer exists.
- For $0 \leq m \leq n$: $[m, 0, \dots, 0, m]$.
- Otherwise, start with array of length n $[n, 0, \dots, 0]$.

K. Mad MAD Sum II

- Then (all arrays are in length n):
 - $[n, \dots, 0, n]$
 - $[n, \dots, n, 0]$
 - ...
 - $[n, n, 0, \dots, 0, 0]$
 - $[n, n, 0, \dots, 0, n]$
 - $[n, n, 0, \dots, n, 0]$
 - ...
 - $[n, n, n, \dots, 0]$
- At each step, MAD increases by n .
- When we are close enough to m , say the difference with m is δ :
 - We had $[n, \dots, n, 0, 0, \dots, n, \dots, 0, 0]$
 - We add δ s: $[n, \dots, n, \delta, 0, \dots, n, 0, \dots, \delta]$

K. Mad MAD Sum II

- In two cases we fail:
 - $[n, \dots, n, 0, \dots, 0]$
 - $[n, \dots, n, 0, \dots, 0, n]$
- $[n, \dots, n, 0, \dots, 0, n]$ can be converted to $[n, \dots, n, 0, \dots, 0]$ and $\text{delta} < 2n$
- Now, $\text{delta} < 2n$ and array is $[n, \dots, n, 0, \dots, 0]$ we put $\text{delta} / 2$: $[n, \dots, n, \text{delta}/2, 0, \dots, 0, \text{delta}/2]$
 - Note that two subarrays have $\text{MAD} = \text{delta} / 2$

L. Enigmatic Game



L. Enigmatic Game

- $b[i]$: Number of j 's with $a[j] \geq i$
- Let's see what happens when a move happens.
- $\text{old_}b[i] = (\text{number of old_}a[j] = i) + (\text{number of old_}a[j] > i)$.
- For all $L \leq i < R$, all $a[j] = i$ has been decreased by 1 so $\text{new_}b[i] = (\text{number of old_}a[j] > i) = \text{old_}b[i+1]$. This is left shift $b[L \dots R]$.
- What about $b[R]$? Since we can choose any $a[j] = R$, the value range of $\text{new_}b[R]$ is $[\text{old_}b[R+1], \text{old_}b[R]]$.
- Generally, doing an operation in $[L, R]$ is to set $b[L] :=$ a value in $[b[R+1], b[R]]$, then left shift $b[L \dots R]$.
- Consider different R (including $+\infty$), the value range of $b[L]$ is $[0, b[L]-1]$, this corresponds to "taking any number (>0) of stones from the pile of stones in $b[L]$ ".
- The game is nim on array b !

M. Closed Paths



- Segment tree
- Sqrt
- Euler tour



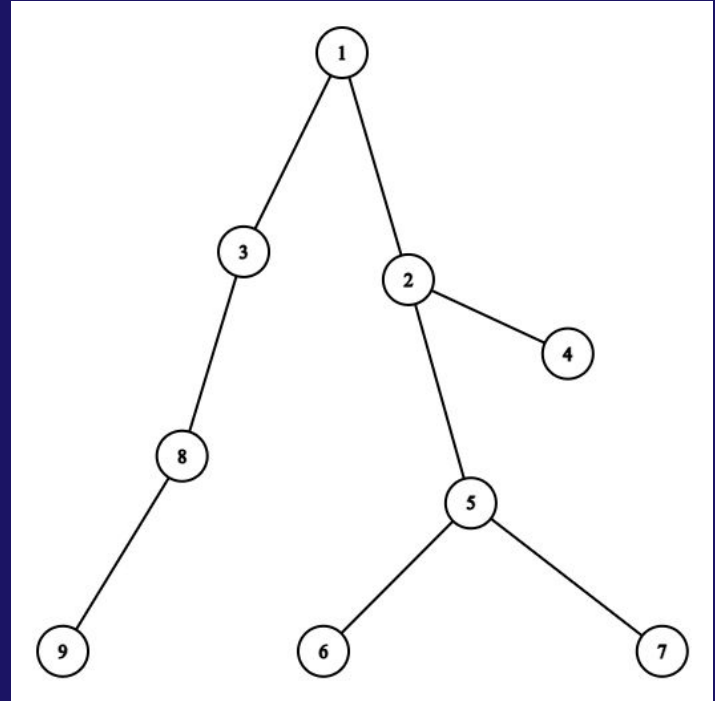
● **Creativity** ● **Knowledge**
● **Implementation**

M. Closed Paths

- The path will be going to a vertex, then repeatedly moving to its parent and it and returning to root.
- ans_v : score if you go down to v , then repeatedly moving to its parent and it and returning to root.
- After transformation, the problem requires two types of modifications and a query:
 - Add value v to the subtree rooted at node x ;
 - Add value v to all child nodes of node x .
 - Find the maximum value.
- Use a variant of the Euler Tour technique.
- First, node 1 is added to the sequence, then $DFS(1)$ is performed. When $DFS(x)$ is called, all child nodes of x are added to the sequence, and then $DFS(son[x][i])$ is performed.
- In this way, the subtree rooted at node x (excluding x itself) and all the child nodes of node x are within a contiguous segment of the sequence. This makes it easy to maintain the operations in $O(n \log n)$.

M. Closed Paths

- DFS order: 1, 3, 8, 9, 2, 5, 6, 7, 4
- BFS order: 1, 3, 2, 8, 5, 4, 9, 6, 7
- DFS-child-first order:
1, 3, 2, 8, 9, 5, 4, 6, 7



M. Closed Paths

- Consider we are updating v .
- Use segment tree on tree to update ans_x for x in subtree of v , except v and its children.
- To update v 's children, use child update query.



Appendix

Nim and Grundy Number

Nim

- In Nim game, if xor sum of all bags is 0, second player wins, otherwise first player wins.

Grundy Number

- Every game can be converted to Nim.
- When some games are separated, use xor to find Grundy number.
- When you want to find Grundy number of a state that can lead to some other states, Grundy number of original state is mex of Grundy number of new states.